

$$X \sim U(-2, 6)$$

a) Trovare la densità PDF di $Y = X - 1$

$$f_Y(y) = f_{X-1}(y) = f_X(y+1)$$

b) Trovare PDF di $Z = |X|$

c) Trovare $E[Z]$

$$f_{X+a}(x) = f_X(x-a)$$

$x(y) = y+1$

d) Trovare $E[XZ]$ e $E[YZ]$

a) $g(x) = x - 1$ $y = g(x) \iff y = x - 1 \iff x = y + 1 \iff x'(y) = 1$

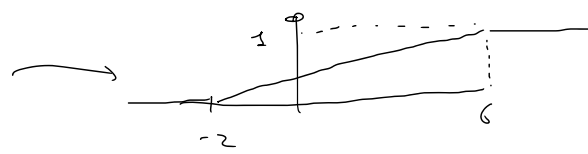
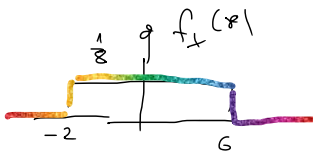
$$f_Y(y) = |x'(y)| f_X(x(y)) = f_X(y+1)$$

Trova la densità di Z

$$= \frac{f_X(x(y))}{|g'(x(y))|}$$

$$g(x) = x - 1 \quad g'(x) = 1 \quad x < -2$$

$$F_Y(x) = \int_{-\infty}^x f_Y(z) dz = \begin{cases} \int_{-2}^x \frac{1}{8} dz = \frac{1}{8}(x+2) & -2 < x < 6 \\ 1 & x > 6 \end{cases}$$



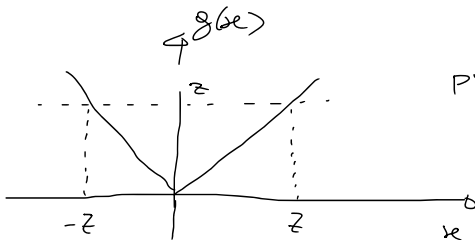
$$F_Y(y) = P[Y \leq y] = P[X - 1 \leq y] = P[X \leq y + 1] = F_X(y + 1)$$

$$f_Y(y) = f_X(y+1)$$

$$Z = |X|$$

$$g(x) = |x|$$

$$f_{X+a}(x) = f_X(x-a)$$



punti $z > 0$

$$f_Z(z) = 0 \quad z < 0$$

$$f_Z(z) = |x_1'(z)| f_X(x_1(z)) + |x_2'(z)| f_X(x_2(z))$$

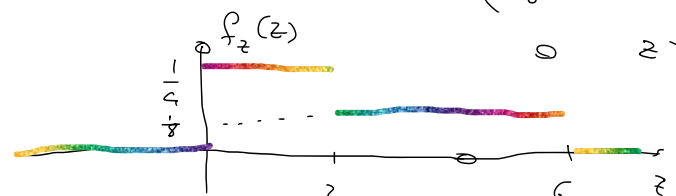
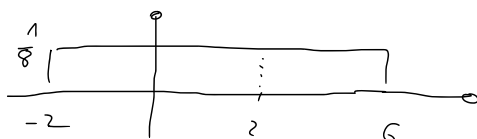
$$x_1(z) = z$$

$$x_1'(z) = 1$$

$$x_2(z) = -z$$

$$x_2'(z) = -1$$

$$f_Z(z) = f_X(z) + f_X(-z) = \begin{cases} \frac{1}{4} & 0 < z < 2 \\ \frac{1}{8} & 2 < z < 6 \\ 0 & z > 6 \end{cases}$$



$$E[Z] = \int_{-\infty}^{+\infty} z f_Z(z) dz = \int_0^2 z \frac{1}{8} dz + \int_2^6 z \frac{1}{4} dz$$

$$E[g(X)] = \int_{-\infty}^{+\infty} g(x) f_X(x) dx = \int_{-2}^6 |x| \frac{1}{8} dx = \int_{-2}^0 \frac{|x|}{8} dx + \int_0^6 \frac{|x|}{8} dx = \int_{-2}^0 -\frac{x}{6} dx + \int_0^6 \frac{x}{8} dx$$

$$E[XZ] = E[Xg(X)] = \int_{-\infty}^{+\infty} x g(x) f_X(x) dx = \int_{-2}^6 x |x| f_X(x) dx = \int_{-2}^0 -\frac{x^2}{8} dx + \int_0^6 \frac{x^2}{8} dx$$

$$E[YZ] = E[(X-1)Z] = E[XZ] - E[Z]$$

$$P[X=k] = A \frac{(\ln(2))^k}{k!} \quad k \in \mathbb{N}_0$$

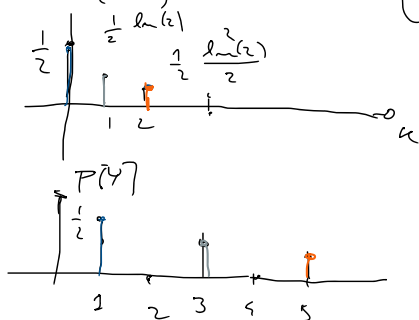
$$e^{-\lambda} \frac{\lambda^k}{k!}$$

$$\lambda = \ln(2)$$

$$A = e^{-\lambda} = e^{-\ln(2)} = e^{\ln(\frac{1}{2})} = \frac{1}{2}$$

$$Y = 2X + 1$$

$$P[Y=k] = P\left[\frac{Y-1}{2} = X\right] = \begin{cases} 0 & k \text{ pari} \\ P[X=\frac{k-1}{2}] = \frac{1}{2} \frac{(\ln(2))^{k-1}}{(\frac{k-1}{2})!} & k \text{ dispari} \end{cases}$$



$$P_i = P[X=x_i] \quad P[X=x_i | Y=y_j] = P[X_1 Y=y_j = x_i]$$

$$q_j = P[Y=y_j]$$

$$P[X=x_i, Y=y_j] = P_i q_j^{(i)} = q_j \cdot P_i^{(j)}$$

$$P[Y=y_j | X=x_i] =$$

$$P[Y=y_j | X=x_i]$$

$$H(X | Y=y_j) = H(X_{|Y=y_j}) = \sum_{i=1}^M P_i^{(j)} \log \frac{1}{P_i^{(j)}}$$

$$H(X|Y) = \sum_{j=1}^M P[Y=y_j] H(X | Y=y_j)$$

$$= \sum_j q_j \sum_i P_i^{(j)} \log \frac{1}{P_i^{(j)}} = \sum_{i,j} P[X=x_i, Y=y_j] \log \frac{1}{P_i^{(j)}}$$

$$H(X) - H(X|Y) = \sum_i P_i \log \frac{1}{P_i} - \sum_i P_i \left(\sum_j q_j^{(i)} \right) \log \frac{1}{P_i} =$$

$$= \sum_{i,j} P[X=x_i, Y=y_j] \log \frac{1}{P_i} - \sum_{i,j} P[X=x_i, Y=y_j] \log \frac{1}{P_i^{(j)}}$$

$$= \sum_{i,j} P[X=x_i, Y=y_j] \log \frac{P_i^{(j)}}{P_i}$$

$$\sum_{i,j} P[X=x_i, Y=y_j] \log_2 \frac{P_{ij}^{(X)}}{P_i} = \sum_{i,j} P[X=x_i, Y=y_j] \log_2 \frac{P_i P[X=x_i, Y=y_j]}{P[X=x_i] P[Y=y_j]}$$

$$\frac{P_{ij}^{(X)}}{P_i} = \frac{P_{ij}^{(X)} q_j}{P_i q_j} = \frac{P[X=x_i, Y=y_j]}{P[X=x_i] P[Y=y_j]}$$

$$H(Y|X=x_i) = H\left(\frac{Y}{X=x_i}\right) = \sum_{j=1}^M q_j^{(i)} \log_2 \frac{1}{q_j^{(i)}}$$

$$H(Y|X) = \sum_{i=1}^M P[X=x_i] H[Y|X=x_i] = \sum_i P_i \sum_j q_j^{(i)} \log_2 \frac{1}{q_j^{(i)}} =$$

$$\sum_{i,j} P_i q_j^{(i)} \log_2 \frac{1}{q_j^{(i)}} =$$

$$= \sum_{i,j} P[X=x_i, Y=y_j] \log_2 \frac{1}{P[Y=y_j|X=x_i]}$$

$$H(Y) - H(Y|X) =$$

$$H(Y) = \sum_{j=1}^M q_j \log_2 \frac{1}{q_j} = \sum_{j=1}^M q_j \left[\sum_{i=1}^M P_i^{(j)} \right] \log_2 \frac{1}{q_j} = \sum_{i,j} q_j P_i^{(j)} \log_2 \frac{1}{q_j}$$

$$= \sum_{i,j} P[X=x_i, Y=y_j] \log_2 \frac{1}{q_j}$$

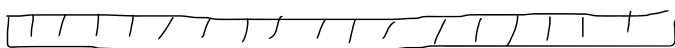
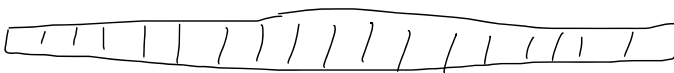
$$= \sum_{i,j} P[X=x_i, Y=y_j] \log_2 \frac{q_j^{(i)}}{q_j}$$

$$\frac{q_j^{(i)}}{q_j} = \frac{P_i q_j^{(i)}}{P_i q_j} = \frac{P[X=x_i, Y=y_j]}{P[X=x_i] P[Y=y_j]}$$

$$= \sum_{i,j} P[X=x_i, Y=y_j] \log_2 \frac{P[X=x_i, Y=y_j]}{P[X=x_i] P[Y=y_j]}$$

$$P_i \sum_i P_i \log_2 \frac{P_i}{q_i}$$

$$q_i$$



}

$X_i(\omega)$

$\left\{ \begin{array}{ll} \text{numerico} & \text{se } \omega \text{ sono fissati} \\ \text{v. a.} & \text{se } \omega \text{ fissati e } i \text{ variabile} \\ \text{numerico} & \text{se } \omega \text{ fissati e } i \text{ variabile} \\ \text{probabilistico} & \text{se } \omega \text{ e } i \text{ variabili} \end{array} \right.$

9
c.

procedi
decisioni
X_i

$$\underline{X}_n = (X_1, X_2, \dots, X_n)$$

$$\underline{x}_n = (x_1, x_2, \dots, x_n)$$

$$\underline{X}_n = \underline{x}_n \Leftrightarrow [X_1 = x_1] \text{ AND } [X_2 = x_2] \text{ AND } \dots \text{ AND } [X_n = x_n]$$

$$P[\underline{X}_n = \underline{x}_n] = P[X_1 = x_1, X_2 = x_2, \dots, X_n = x_n] \stackrel{\text{ind. p.}}{=} P[X_1 = x_1] P[X_2 = x_2] \dots P[X_n = x_n] = \prod_{i=1}^n P[X_i = x_i]$$

$$H(\underline{X}_n) = \sum_{\underline{x}_n \in \mathcal{X}^n} P[\underline{X}_n = \underline{x}_n] \log_2 \frac{1}{P[\underline{X}_n = \underline{x}_n]}$$

$$= \sum_{x_1 \in \mathcal{X}} \sum_{x_2 \in \mathcal{X}} \dots \sum_{x_n \in \mathcal{X}} P[X_1 = x_1] \cdot P[X_2 = x_2] \dots P[X_n = x_n] \left[\log_2 \frac{1}{P[X_1 = x_1]} + \log_2 \frac{1}{P[X_2 = x_2]} + \dots + \log_2 \frac{1}{P[X_n = x_n]} \right]$$

$$= \sum_{x_1} \sum_{x_2} \dots \sum_{x_n} P[X_1 = x_1] P[X_2 = x_2] \dots P[X_n = x_n] \log_2 \frac{1}{P[X_1 = x_1]}$$

$$+ \dots = \log_2 \frac{1}{P[X_2 = x_2]} = \log_2 \frac{1}{P[X_3 = x_3]} = \dots = \log_2 \frac{1}{P[X_n = x_n]}$$

$$= \left(\sum_{x_1} P[X_1 = x_1] \log_2 \frac{1}{P[X_1 = x_1]} \right) \left(\sum_{x_2} P[X_2 = x_2] \right) \dots \left(\sum_{x_n} P[X_n = x_n] \right)$$

$$+ \left(\sum_{x_2} P[X_2 = x_2] \log_2 \frac{1}{P[X_2 = x_2]} \right) \left(\sum_{x_1} P[X_1 = x_1] \right) \left(\sum_{x_n} P[X_n = x_n] \right)$$

$$+ \left(\sum_{x_n} P[X_n = x_n] \log_2 \frac{1}{P[X_n = x_n]} \right) \left(\sum_{x_1} P[X_1 = x_1] \right) \left(\sum_{x_2} P[X_2 = x_2] \right) \dots \left(\sum_{x_{n-1}} P[X_{n-1} = x_{n-1}] \right)$$

$$= H(X_1) + H(X_2) + \dots + H(X_n) = n H(X_1)$$

$$\sum_{x_1, x_2} \underbrace{P(x_1) P(x_2^{(x_1)})}_{P(x_1, x_2)} \log \frac{1}{\underbrace{P(x_2^{(x_1)})}_{P(x_2 | x_1)}} = H(X_2 | X_1)$$

$$\sum_{x_1, x_2, x_3} \underbrace{P(x_1) P(x_2^{(x_1)}) P(x_3^{(x_1, x_2)})}_{P(x_1, x_2, x_3)} \log \frac{1}{\underbrace{P(x_3^{(x_1, x_2)})}_{P(x_3 | x_1, x_2)}} = H(X_3 | X_1, X_2)$$